On some spectral spaces associated to tensor triangular categories

Abhishek Banerjee

Indian Institute of Science, Bangalore

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- We will give examples of spectral spaces from categorical algebra

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• A space X is spectral if and only if X is T_0 and there is a subbasis $\mathcal{S}$ of X such that for any ultrafilter $\mathfrak{F}$ on X, the set

$$X_{\mathcal{S}}(\mathfrak{F}) = \{x \in X \mid \forall S \in \mathcal{S}, \text{ we have } x \in S \Leftrightarrow S \in \mathfrak{F}\}$$

is non-empty.

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• Finocchiaro, Fontana, Spirito used it to give examples from commutative algebra

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• For $Q \in Fin(N, M)$, we consider:

$$V(Q, N, M) := \{T \in Fin(N, M) \mid Q \subseteq T\}$$

$$D(Q, N, M) := Fin(N, M) \backslash V(Q, N, M)$$

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- "Test element" for ultrafilter $\mathfrak{F}$:

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Modules over tensor triangular categories

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- (Balmer) A thick tensor ideal $\mathcal{P} \subsetneq \mathcal{K}$ is prime if $x \otimes y \in \mathcal{P}$ for some x, $y \in \mathcal{K}$ means that at least one of x and y is in $\mathcal{P}$.

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- Balmer began the study of tensor triangular geometry by constructing the space $Spec(\mathcal{K})$ of prime ideals in $(\mathcal{K}, \otimes, 1)$.

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A full subcategory $\mathcal{L} \subseteq \mathcal{M}$ containing 0 is said to be a thick $\mathcal{K}$-submodule of $\mathcal{M}$ if it satisfies the following three conditions:

(a) The composition of functors

$$\mathcal{K} \times \mathcal{L} \hookrightarrow \mathcal{K} \times \mathcal{M} \xrightarrow{*} \mathcal{M} \tag{1}$$

factors through $\mathcal{L}$.

(b) Given objects m, $m' \in \mathcal{M}$, then $m \oplus m' \in \mathcal{L}$ if and only if both m, $m' \in \mathcal{L}$.

(c) For any distinguished triangle $m' \longrightarrow m \longrightarrow m''$ in $\mathcal{M}$, if two out of the three objects m', m, m'' lie in $\mathcal{L}$, so does the third.

The collection of all thick $\mathcal{K}$-submodules of $\mathcal{M}$ will be denoted by $SMod(\mathcal{M})$.

Topology on $SMod(\mathcal{M})$

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(a) $U(0) = SMod(\mathcal{M})$.

(b) For any distinguished triangle $m' \longrightarrow m \longrightarrow m''$ in $\mathcal{M}$, we have $U(m) \supseteq U(m') \cap U(m'')$.

(c) For objects $m,\ m' \in \mathcal{M}$, we have $U(m \oplus m') = U(m) \cap U(m')$.

(d) For any $a \in \mathcal{K}$ and any $m \in \mathcal{M}$, we have $U(m) \subseteq U(a * m)$.

(e) If $T_{\mathcal{M}}$ is the translation functor on $\mathcal{M}$, we have $U(T_{\mathcal{M}}(m)) = U(m)$ for each $m \in \mathcal{M}$.

An operator

$$c : SMod(\mathcal{M}) \longrightarrow SMod(\mathcal{M})$$

(a) Extensive if $\mathcal{N} \subseteq c(\mathcal{N})$ for each $\mathcal{N} \in SMod(\mathcal{M})$.

(b) Order-preserving if $\mathcal{N} \subseteq \mathcal{N}'$ implies that $c(\mathcal{N}) \subseteq c(\mathcal{N}')$.

(c) Idempotent if $c(\mathcal{N}) = c(c(\mathcal{N}))$ for each $\mathcal{N} \in SMod(\mathcal{M})$.

(d) Finite type if $c(\mathcal{N}) = \bigcup\limits_{n \in \mathcal{N}} c(\mathcal{K}(n))$, where $\mathcal{K}(n)$ is the smallest thick $\mathcal{K}$ submodule of $\mathcal{M}$ containing n.

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Theorem: $SMod^c(\mathcal{M})$ of fixed points of c is a spectral space having the collection $\{U(m) \cap SMod^c(\mathcal{M})\}_{m\in\mathcal{M}}$ as a basis of quasi-compact open subspaces.

• (Balmer) There is a bijection between radical thick tensor ideals in $(\mathbb{T}, \otimes, 1)$ and open subspaces in the inverse topology on $Spec(\mathbb{T})$.

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• (Finocchiaro, Fontana, Spirito) Let R be a commutative ring. Then, there is a homeomorphism between the space of non-empty closed subspaces of $Spec(R)$ and the inverse topology on the spectral space of proper radical ideals of R.

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(1) the spectral space of radical thick tensor ideals in $(\mathbb{T}, \otimes, \mathbf{1})$ *with topology generated by the open sets*

$$\{\mathcal{K} \in Id(\mathbb{T}) \mid \mathcal{K} \text{ is radical and } a \notin \mathcal{K}\} \qquad \forall\, a \in \mathbb{T}$$

(2) the space $\mathcal{U}(\mathrm{Spec}(\mathbb{T})^{\mathrm{inv}})$ *of open sets of* $\mathrm{Spec}(\mathbb{T})^{\mathrm{inv}}$ *with topology generated by taking the collection*

$$\{V \in \mathcal{U}(\mathrm{Spec}(\mathbb{T})^{\mathrm{inv}}) \mid V \not\supseteq U\} \qquad \forall\, U \in \mathcal{U}(\mathrm{Spec}(\mathbb{T})^{\mathrm{inv}})$$

to be open sets.